

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 917+050

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – VUP

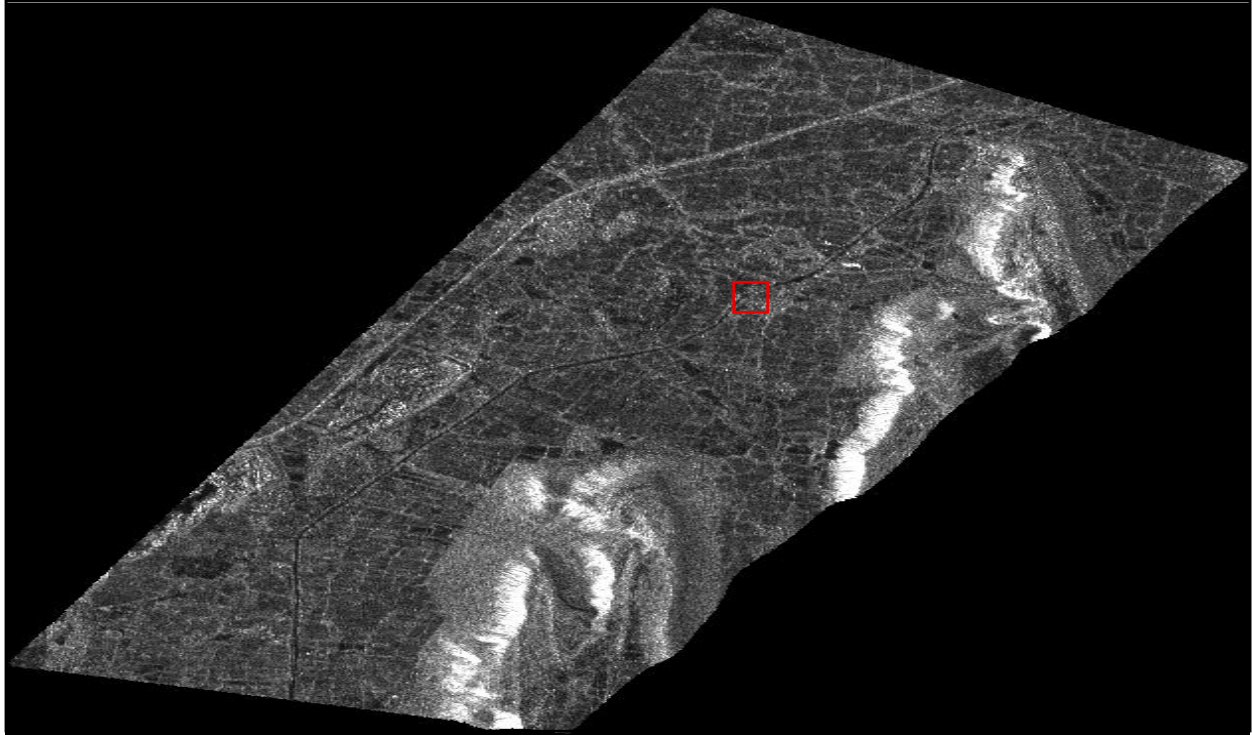
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

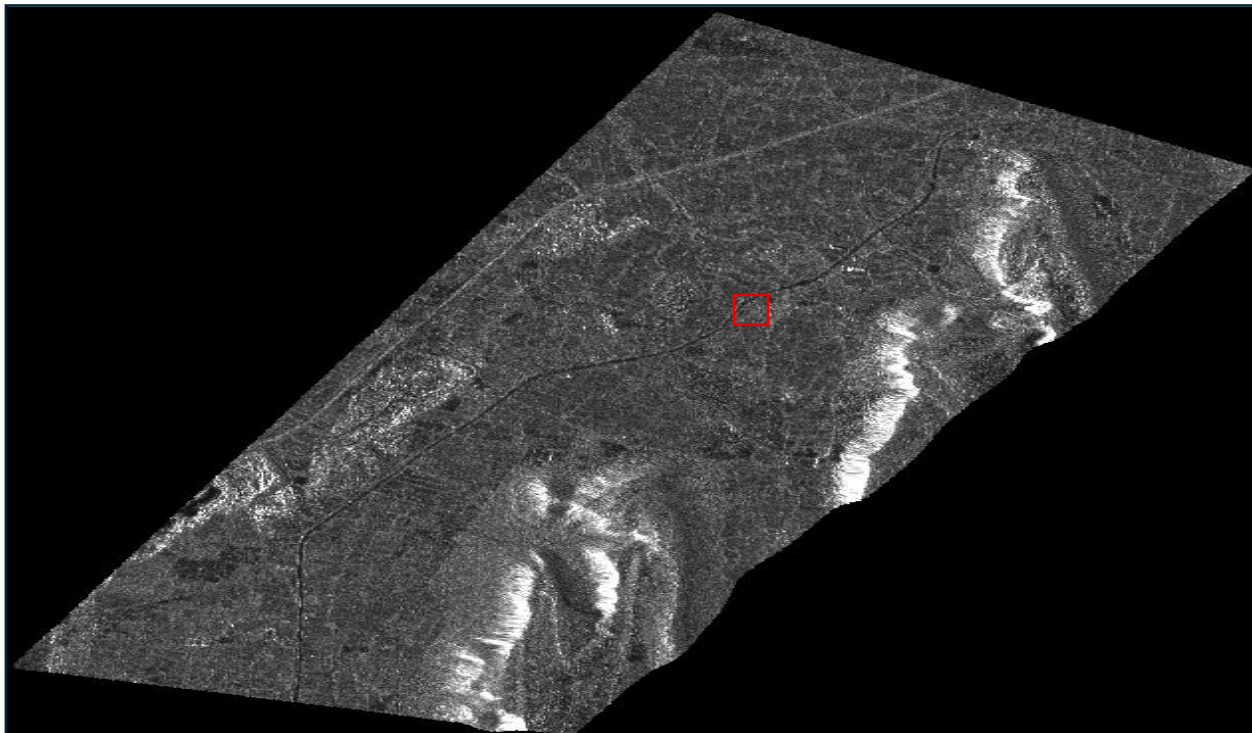
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

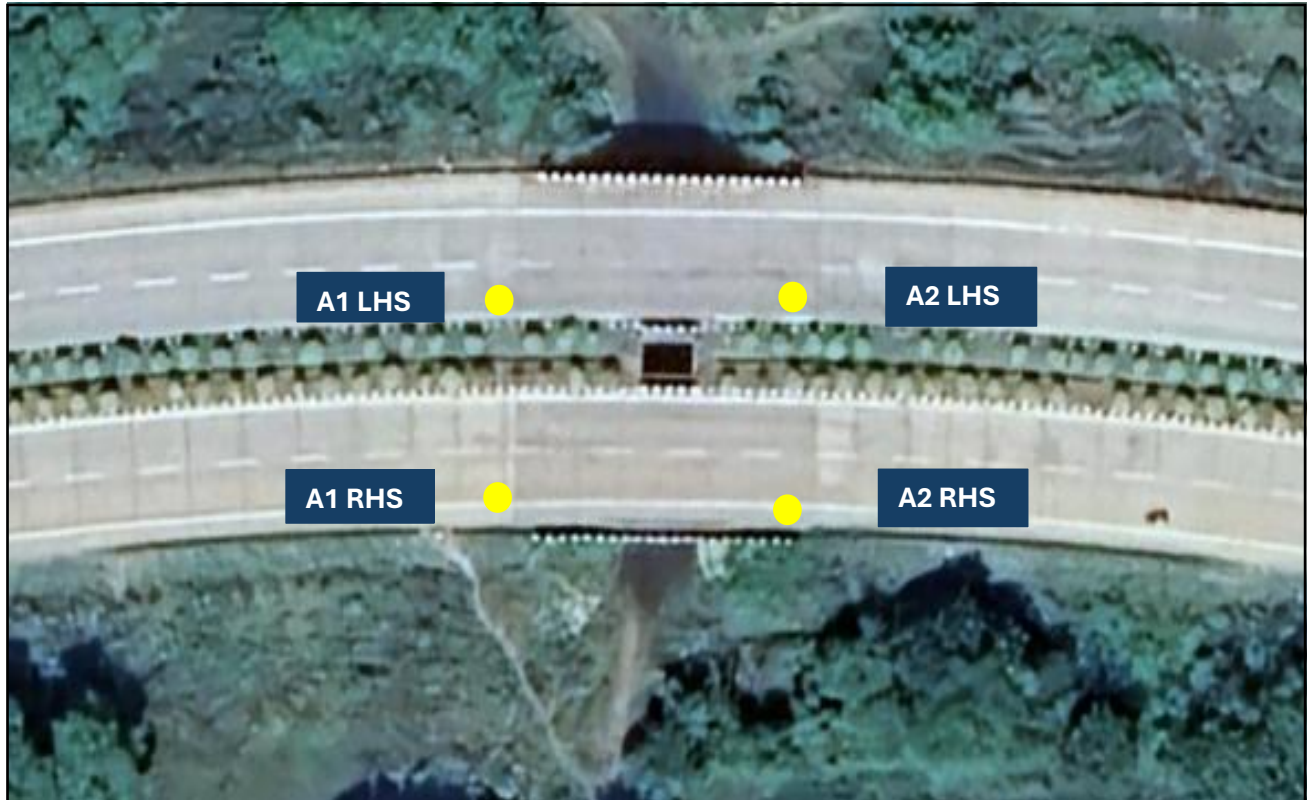


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

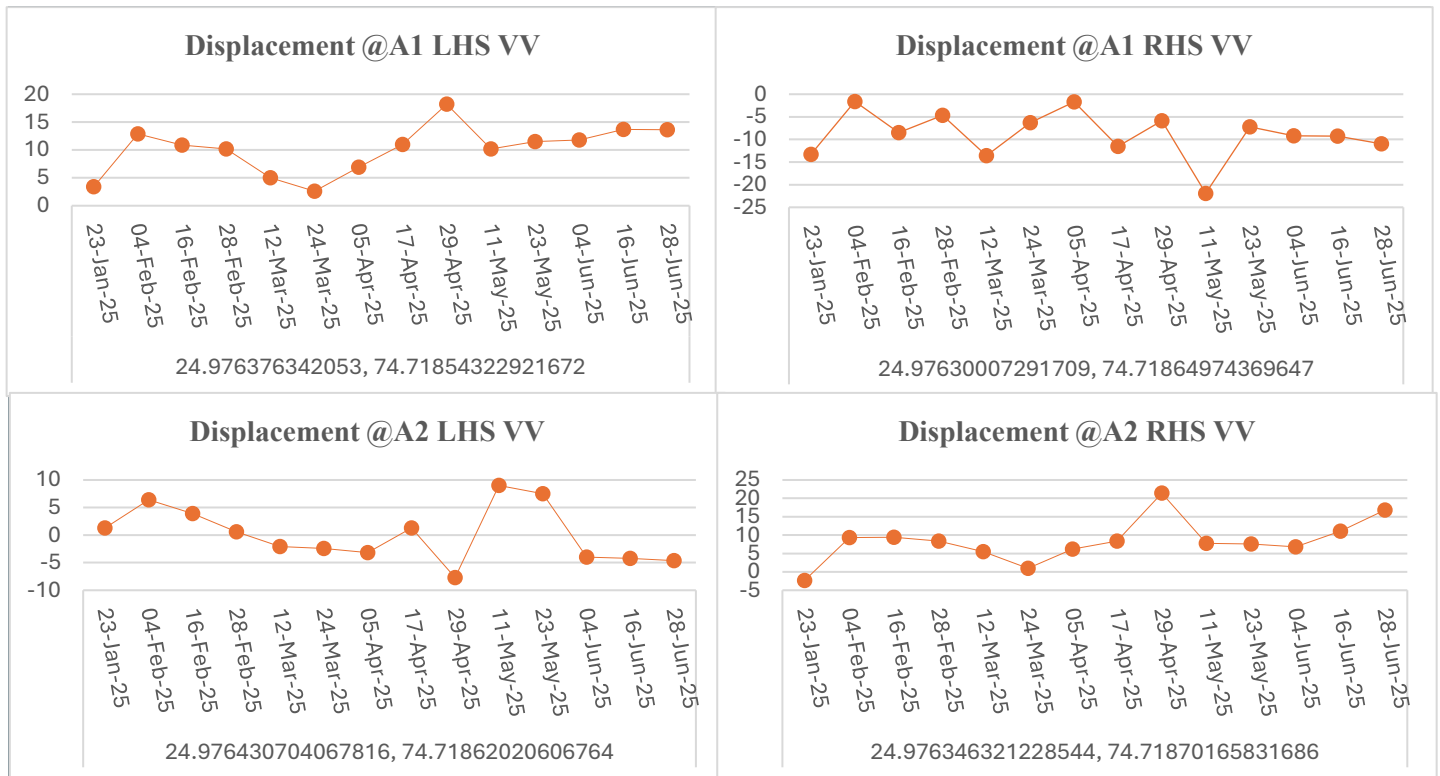
VUP- CH-917+050

Length – 22.26M

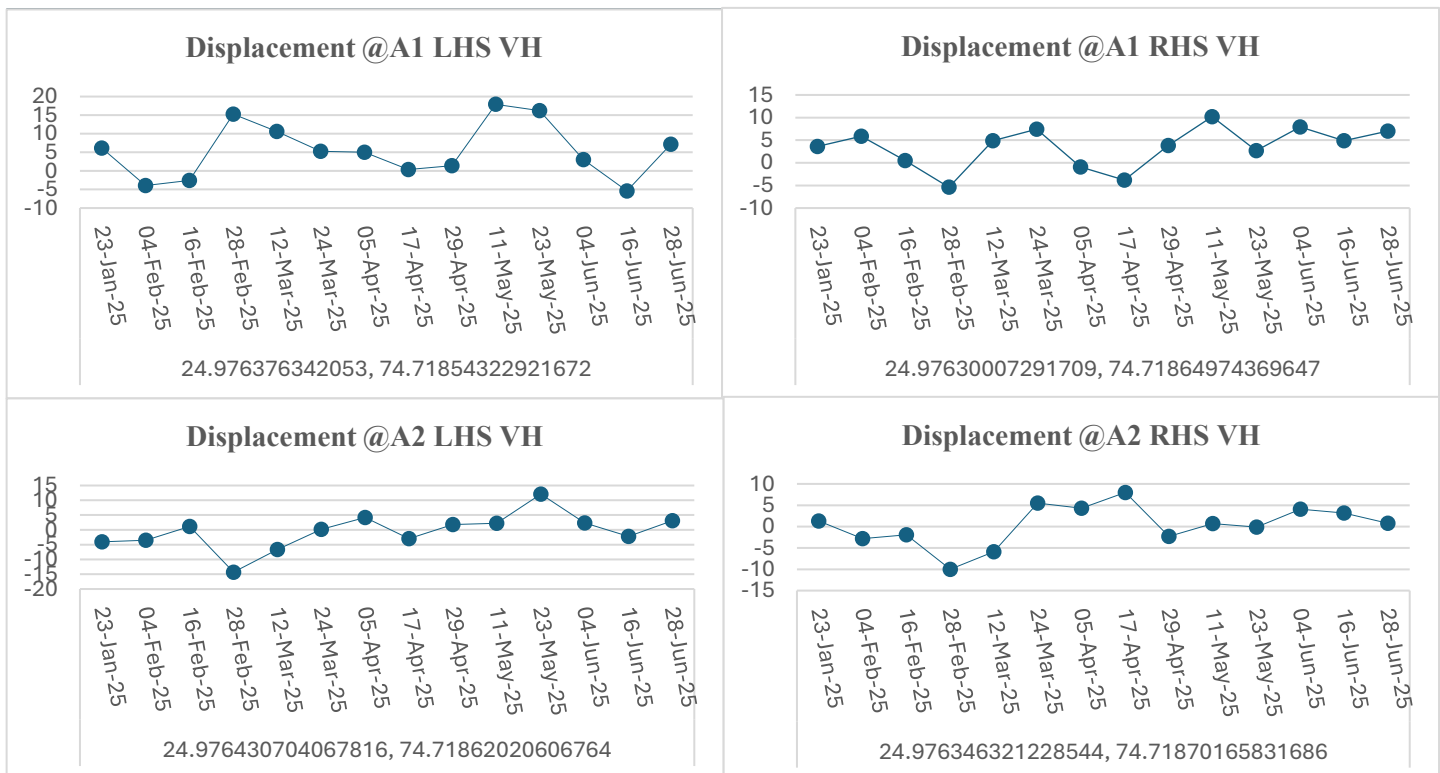


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



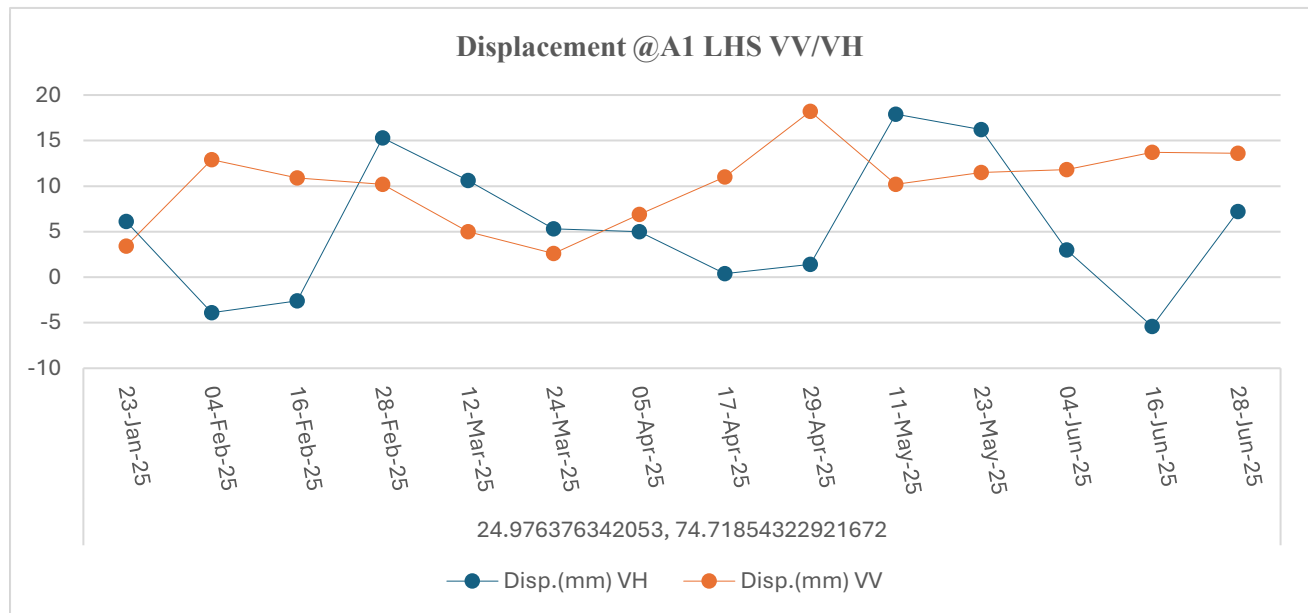
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.976376342053, 74.71854322921672)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** exhibits moderate variability with a predominantly positive trend, indicating overall vertical uplift at the A1 LHS location during most of the monitoring period.
- **Horizontal Displacement (VH)** shows high variability with alternating positive and negative phases, ranging roughly between –5 mm and +18 mm.

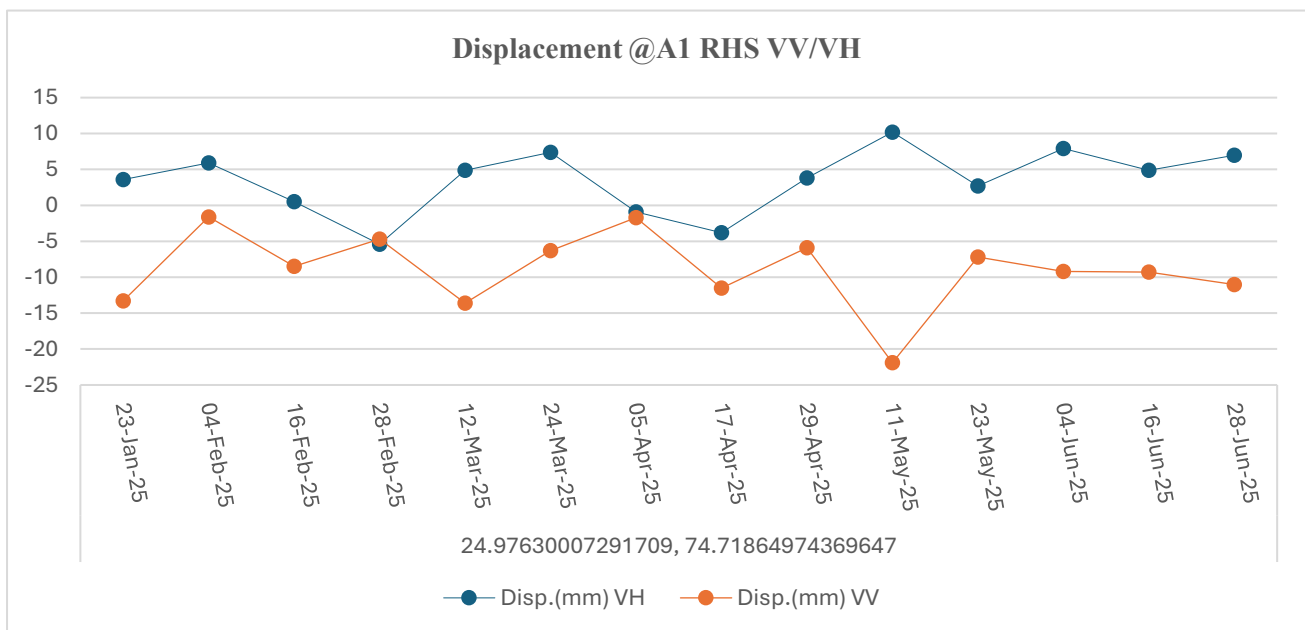
Critical Events & Anomalies:

- **January 2025:** VH begins near +6 mm, while VV is approximately +4 mm, indicating initial coupled uplift in both components.
- **May 2025:** VH remains elevated (+16 mm), while VV stabilises near +11 mm, indicating sustained coupled uplift.
- **June 2025:** VH recovers to +7 mm, while VV remains elevated (+13 mm), indicating vertical stability with partial lateral recovery by the end of monitoring.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.97630007291709, 74.71864974369647)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows moderate variability with predominantly positive values, ranging roughly between -4 mm and +10 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits strong negative dominance with high variability, indicating prevailing vertical settlement at the A1 RHS location over the monitoring period.

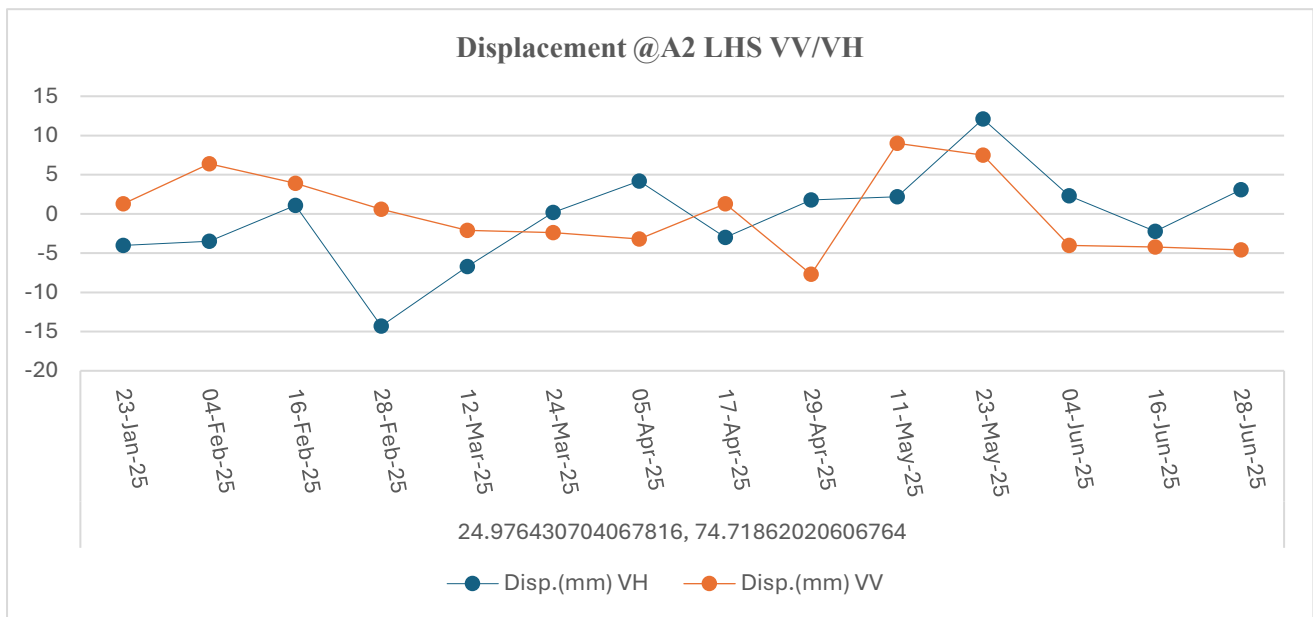
Critical Events & Anomalies:

- **January 2025:** VH begins near +4 mm, while VV is around -13 mm, indicating early vertical settlement accompanied by mild lateral displacement.
- **April 2025:** VH reaches -4 mm, while VV remains negative (-6 mm), marking a temporary lateral reversal under ongoing settlement.
- **June 2025:** VH stabilises between +5 and +7 mm, while VV remains consistently negative indicating no sustained vertical recovery by the end of monitoring.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.976430704067816, 74.71862020606764)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with both positive and negative excursions, ranging roughly between +12 mm and –15 mm.
- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with alternating uplift and settlement phases, indicating a mixed vertical response at the A2 LHS location.

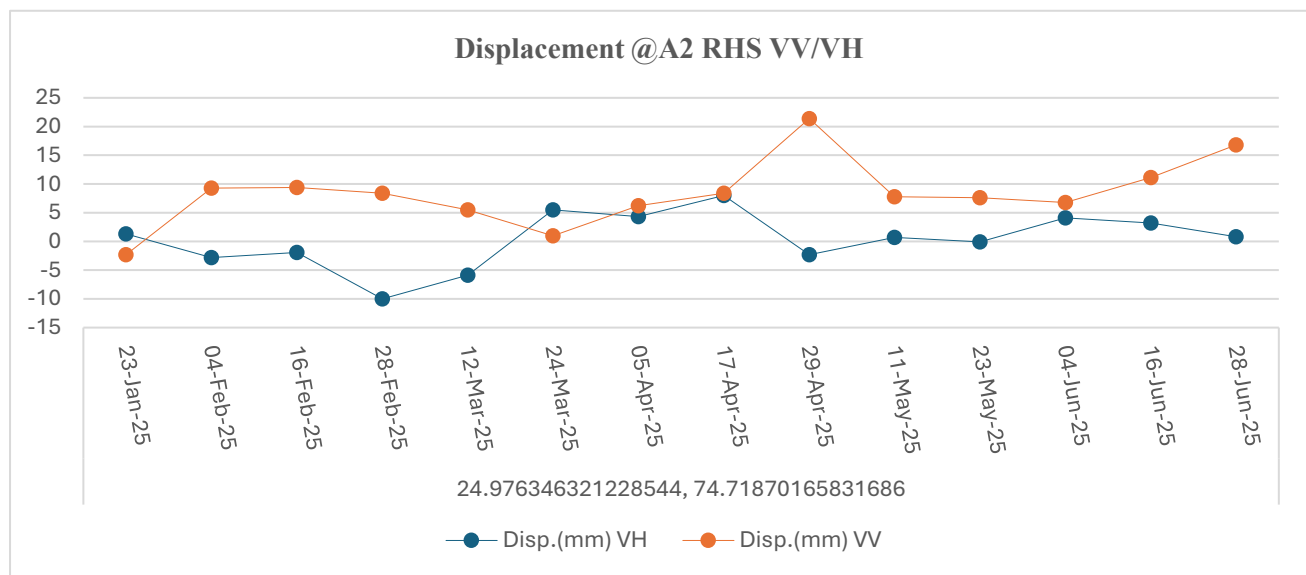
Critical Events & Anomalies:

- **January 2025:** VH starts near –4 mm, while VV is slightly positive (+1 mm), indicating initial lateral settlement with mild vertical uplift.
- **June 2025:** VH recovers slightly to +3 mm, while VV remains negative (–4 mm), indicating no full vertical recovery by the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.976346321228544, 74.71870165831686)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement shows predominantly positive values throughout the monitoring period, indicating a dominant uplift behaviour at the A2 RHS location.
- **Horizontal (VH) displacement** Horizontal displacement exhibits moderate to high variability, oscillating between approximately +6 mm and –10 mm.

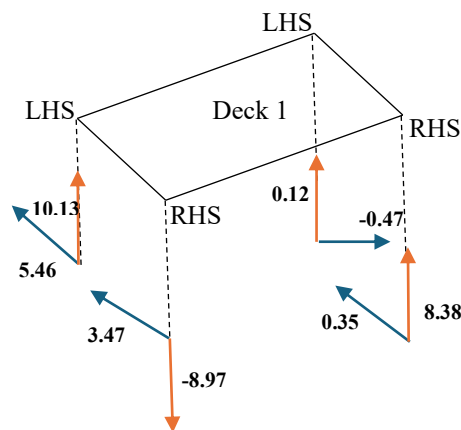
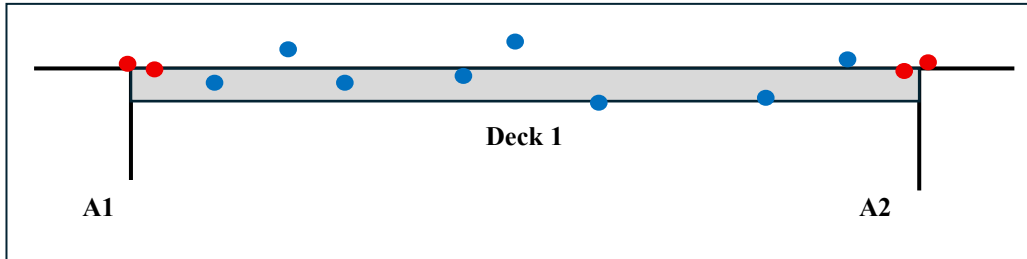
Critical Events & Anomalies:

- **January:** VH starts near +2 mm, while VV is slightly negative (–1 mm), indicating minor lateral uplift with marginal vertical settlement.
- **May 2025:** VH stabilises near 0 mm, while VV reduces to +7 mm, suggesting partial relaxation after the peak uplift.
- **June 2025:** VH decreases to 2 mm, while VV reaches +16 mm, indicating strong vertical uplift persisting through the end of the monitoring period.

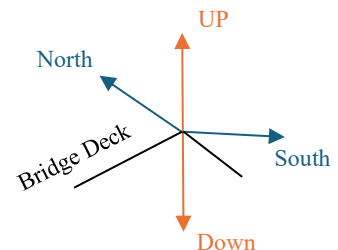
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The maximum downward settlement is observed at the central RHS lower region, reaching approximately -8.97 mm, identifying this as the most vertically critical location. The coexistence of uplift at both supports and settlement at the central lower region confirms vertical rotation combined with localized bending, rather than uniform vertical deflection.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability, reflecting torsional deformation behaviour of the deck slab. The contrast in magnitude and direction of horizontal displacements across the deck confirms torsional rotation about a longitudinal axis, rather than rigid-body translation

CONCLUSION:

- The bridge deck slab exhibits a torsion-influenced deformation pattern, characterized by asymmetric vertical uplift and settlement combined with differential horizontal displacement. Overall, the observed displacement behaviour is moderate but structurally relevant, warranting continued monitoring and correlation with loading conditions and support performance.